

15. Let $A = \begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. Show that the equation $Ax=b$ does not have a solution for all possible b , and describe the set of all b for which $Ax=b$ does have a solution.

Solution: By Theorem 3, the solution of the equation $Ax=b$ is the same as that of the system of linear equations corresponding to the augmented matrix $\begin{bmatrix} a_1 & a_2 & b \end{bmatrix}$.

$$\begin{bmatrix} 2 & -1 & b_1 \\ -6 & 3 & b_2 \end{bmatrix} \xrightarrow[\text{+3xRow 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 2 & -1 & b_1 \\ 0 & 0 & 3b_1 + b_2 \end{bmatrix}$$

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Verification of whether the final matrix is in echelon form

1. If $3b_1 + b_2 = 0$, Row 2 is a zero row. Row 1 is non-zero. $\therefore$ The condition that all non-zero rows must be above all zero rows is True.
2. If $3b_1 + b_2 \neq 0$, then both Row 1 and Row 2 are non-zero. $\therefore$ The condition that all non-zero rows must be above all zero rows is vacuously True.
3. Row 1 Leading entry = 2 in Column 1
Row 2 Possible leading entry = $3b_1 + b_2$ in Column 3
 $\therefore$ The condition that for each row, its leading entry is in a column to the right of the columns of the leading entries of all rows above it is True.
3. Row 2 Column 1 = 0
 $\therefore$ The condition that for each pivot column, the elements below the pivot position must be all 0 is True.
- $\therefore$ The matrix $\begin{bmatrix} 2 & -1 & b_1 \\ 0 & 0 & 3b_1 + b_2 \end{bmatrix}$ is in echelon form. $\square$

There is a row $\begin{bmatrix} 0 & 0 & 3b_1 + b_2 \end{bmatrix}$. By Theorem 2, the system of equations corresponding to the augmented matrix is consistent, i.e. there exists a solution set iff $3b_1 + b_2 = 0$.